Reviewer Pack — Minimal Local Cohomology Bounds Communication Complexity

Entry 4c97cf293a35 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Cohomology Bounds Communication Complexity

Entry ID: `4c97cf293a35` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-29 20:48:37 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Local Cohomology) **Field B** (complexity object): Boolean Function Complexity (Communication Complexity)

Statement:

> For a given Boolean function f with n variables, the minimal local cohomology degree of the variety defined by its negation, denoted as h(f), is upper bounded by the communication complexity of f, i.e., h(f) ≤ CC(f). Furthermore, for all instances with property P, property Q holds: if f has a communication protocol with k rounds, then h(f) = O(k).

Rationale (proposer's reasoning):

> Local cohomology in algebraic geometry can capture geometric properties that are relevant to the complexity of Boolean functions, especially when considering their representations as varieties. By mapping instances of Boolean functions into this algebraic framework, we might uncover new insights into communication complexity.

Taxonomy category: `cg_kw_andreev` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `6a61f36653611d0e`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> If for all 30 seeds, the Pearson correlation coefficient between minimal local cohomology degree h(f) and communication complexity CC(f) is greater than or equal to 0.7, then the conjecture is supported.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "local cohomology" AND "communication complexity" - "minimal local cohomology degree" AND Boolean function - "property P and property Q" AND communication protocol

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction

def generate_boolean_function(n):
    return [random.choice([0, 1]) for _ in range(2**n)]

def communication_complexity(f):
    n = int(math.log2(len(f)))
    if len(f) != 2**n:
        raise ValueError("Input must be a Boolean function with 2^n variables")

    def simulate_protocol(k):
        new_f = [f[i] ^ f[i + 2**(k-1)] for i in range(2**(n-k))]
        return any(new_f)

    k = 1
    while True:
        if not simulate_protocol(k):
            break
        k += 1
    return k

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [10, 15, 20, 25]
    results = []

    for n in n_values:
        f = generate_boolean_function(n)
        cc = communication_complexity(f)
        h_f = Fraction(1, 2**n)  # Simplified local cohomology degree
        results.append({"n": n, "cc": cc, "h_f": h_f})
```

```
    metric_name = "Communication Complexity vs Local Cohomology"
    metric_value = sum(result["cc"] * result["h_f"] for result in results) / len(results)
    instances_tested = len(results)
    n_max = max(result["n"] for result in results)
    conjecture_holds = all(result["cc"] >= result["h_f"] for result in results)
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": metric_name,
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value)**2 for r in results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_fai
    else:
        print(f"RESULT: INCONCLUSIVE reason=unknown")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb42f021.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb42f021.py", line 44, in run_trial
    cc = communication_complexity(f)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb42f021.py", line 32, in communication
    if not simulate_protocol(k):
           ^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_eb42f021.py", line 27, in simulate_prot
    new_f = [f[i] ^ f[i + 2**(k-1)] for i in range(2**(n-k))]
                                            ^^^^^^^^^^^^^^^^
TypeError: 'float' object cannot be interpreted as an integer
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test code crashed during execution, which prevented the production of data necessary to evaluate the conjecture. | next: Investigate and fix the error in the test code that caused it to crash. Once the code is corrected, rerun the test with a valid dataset to assess the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17185
2	propose	ollama_remote	glm4:latest	0	0	15982
3	propose	ollama_remote	glm4:latest	0	0	13868
4	preregistration	ollama_remote	glm4:latest	0	0	10795
5	novelty	ollama_remote	glm4:latest	0	0	8457
6	novelty	ollama_remote	glm4:latest	0	0	14624
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14238
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12251
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15893
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8671
11	judge	ollama_remote	glm4:latest	0	0	12096

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 144059 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/4c97cf293a35.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/4c97cf293a35.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/4c97cf293a35.tar.gz` (if generated)